

SASAKI Shuko

Portfolio: shoqo.github.io

Email: shuko@suou.waseda.jp
Github: github.com/shoqo

EDUCATION

- Waseda University - School of Political Science and Economics Tokyo, Japan
B.A. in Global Political Economy - English-based Degree Program September 2023 - Present

SKILLS

- Programming Languages: Rust, Python, R, JavaScript, C, C++
- Tools: SQL, Google Cloud, Docker, Tableau, Power BI, KNIME
- Statistical Methods: Causal Inference; Matching, Regression Analysis; Instrumental Variables (IV); Regression Discontinuity Design (RDD); Difference-in-Differences (DiD); Conjoint Analysis
- Languages: English (Native); Japanese (Native); Mandarin Chinese (Native)

EXPERIENCE

- Veta Tokyo, Japan (Hybrid)
SWE (Internship) October 2025 - Present
 - Dashboard: Built a client-side administrative panel with integrated visual analytics, including dynamic graphs and heatmaps.
 - Chatbot Integration: Developed and integrated conjoint survey response functionality across LINE, Slack, and Microsoft Teams chatbots, integrating server-side webhooks. Currently implementing an LLM-powered conversational interface that enables users to complete surveys through natural language interactions.
- Japan Biographical Database Tokyo, Japan (Hybrid)
Developer April 2024 - Present
 - 3D Relationship Graph: Developed a 3D relationship graph visualization tool using Three.js to represent complex relationship data.
 - Tech Stack: JavaScript, 3D Force-Directed Graph, AngularJS
- Matsuo Institute Tokyo, Japan (Hybrid)
Research (Internship) August 2025 - March 2026
 - Multi-object Tracking: Implemented YOLOX object detection integrated with Norfair for real-time tracking.
 - Knowledge Tracing: Run experiments on graph-based knowledge tracing models and designed new model structures to improve predictive performance.
- Matsuo-Iwasawa Lab UTokyo and DeepLearning.AI Tokyo, Japan (Remote)
Translator September 2023 - March 2024
 - MOOC translation: Engaged in the translation and localization of an English MOOC on how to develop programs using the OpenAI API and LangChain into Japanese.
- Matsuo Institute Tokyo, Japan (Remote)
Presales (Internship) January 2023 - May 2023
 - Prototyping & UI Design: Developed interactive UI prototypes to demonstrate solution concepts, enabling clients to visualize functionality and provide early feedback.
 - Research & Technical Insights: Surveyed academic papers and industry research to identify potential solutions and provide evidence-based recommendations to clients.

PROJECTS

- YOLOX-Norfair Multi-Object Detection and Tracking: Built a system for multi-object detection and tracking in videos by leveraging YOLOX for object detection and processing the outputs to feed into Norfair for accurate tracking across frames. Technologies used: Python, YOLOX, Norfair.
- 3D Relationship Graph Visualization Tool: Developed an interactive 3D relationship graph visualization tool to represent complex relationship data from the Japan Biographical Database. The tool allows users to explore and analyze relationships in a dynamic 3D space, enhancing data comprehension and user engagement.
- Black-Scholes Implied Volatility and Pnl Visualisation: Project to model and analyze European call and put options using the Black-Scholes pricing framework. The project calculates option prices and evaluates profit and loss (PnL) based on purchase prices, providing a clear view of potential financial outcomes. It includes a dynamic 3D surface visualization that maps PnL against volatility and moneyness, offering insights into risk exposure and payoff profiles under different market conditions.
- Autonomous FPV Drone (In Progress): Building a custom FPV drone with an onboard autonomy stack for point-to-point navigation and real-time collision avoidance. Implementing visual-inertial state estimation, path planning, and obstacle-avoidance control on constrained companion compute, integrating flight controller, sensors, and perception software into a single autonomous system.

PUBLICATIONS

Maurya, S. K., Sasaki, S., Kawano, M., Iwasawa, Y., & Matsuo, Y. *Implicit Graph Learning with Adaptive Temporal Bias for Knowledge Tracing*. Proceedings of the 40th Annual Conference of the Japanese Society for Artificial Intelligence (JSAI 2026), June 2026.